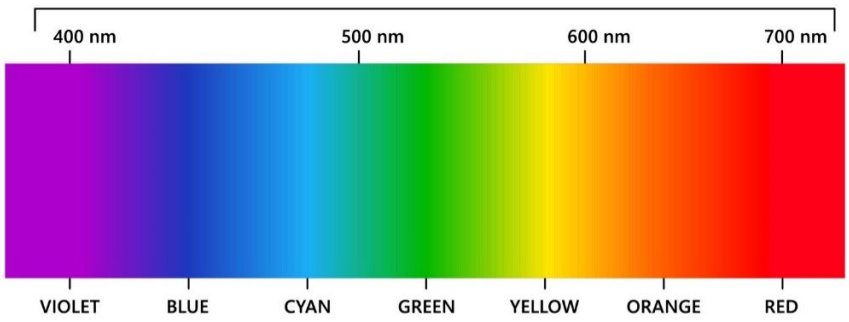
ECE321 Equation Sheet

Semiconductor Physics		
Planck's Constant (h)	6.626×10^{-34} Js	4.141×10^{-15} eVs
Boltzmann's Constant (k)	1.38×10^{-23} J/K	8.62×10^{-5} eV/K
Speed of light (c)	2.998×10^{8} m/s	
Carrier charge (q)	1.6×10^{-19} C	
Intrinsic Carrier Concentration (n_i) (carriers/cm^3)	$B \times T^{3/2} \times e^{-E_g/(2kT)}$	$K = C^{o} + 273.15$ $B_{Si} = 7.3 \times 10^{15}\ cm^{-3}K^{3/2}$ $E_{g,Si} = 1.12\,eV$ At 300K $n_{i,Si} = 1.5 \times 10^{10}\,cm^{-3}$
Carrier Concentration Product	$n \times p = n_i^2$	
Equilibrium Carrier Concentration n-type material (cm^{-3})	$n_{no} \cong N_D$ $p_{no} = n_i^2 / N_D$	
Equilibrium Carrier Concentration p-type material (cm^{-3})	$p_{po} \cong N_A$ $n_{po} = n_i^2 / N_A$	
Energy	$E = \frac{h \times c}{\lambda}$	$1eV = 1.6 \times 10^{-19}\,J$ $E_{eV} = \frac{1240\ eV \cdot nm}{\lambda_{nm}}$
Voltage and Electric Field for constant E-field	$E_{V/cm} = \frac{V}{L}$	

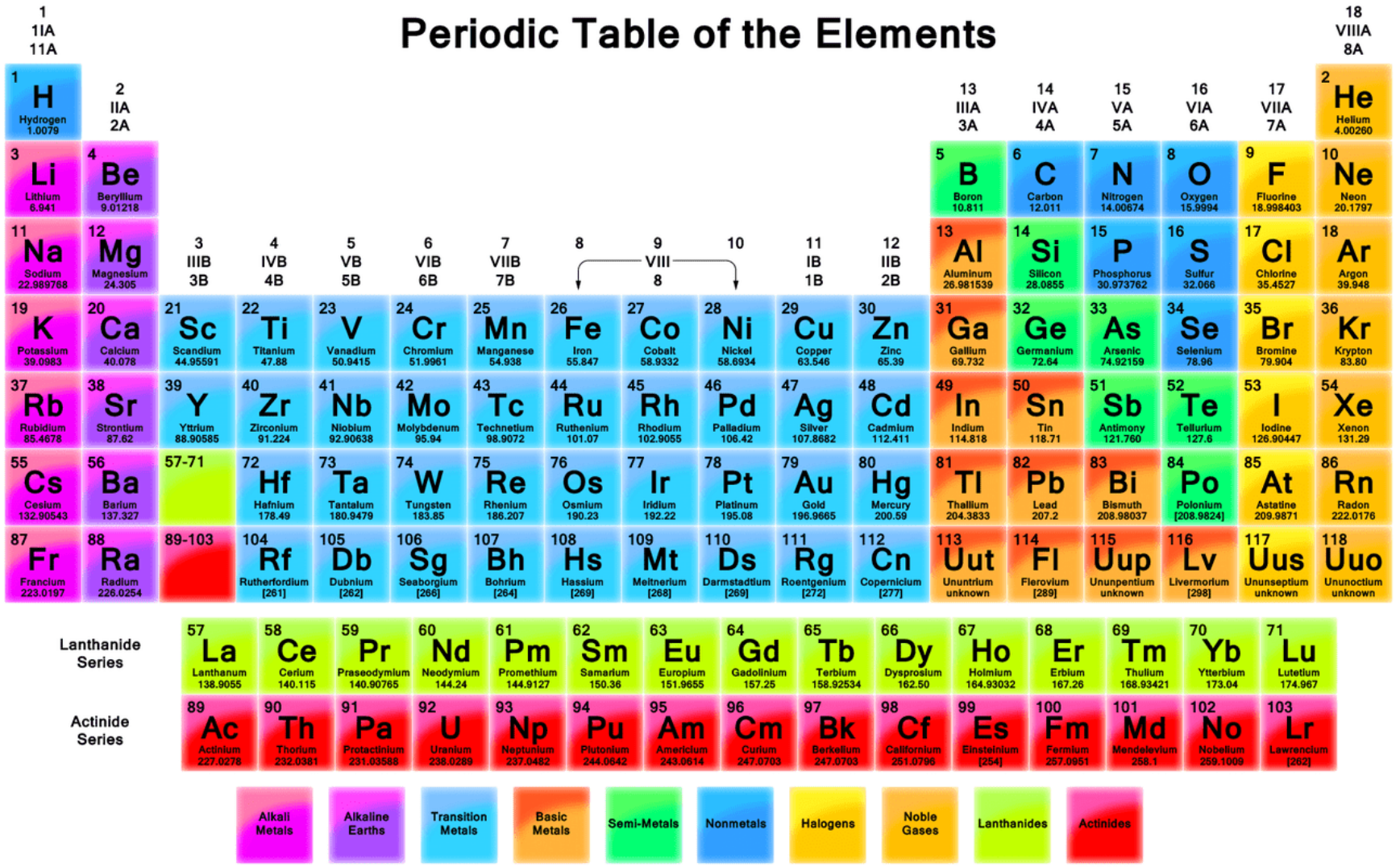
Material	Bandgap Energy E_g (eV)	Mobility μ_n μ_p (cm^2 V^{-1} s^{-1})	Diffusion D_n D_p (cm^2 s^{-1})	Atoms/cm^3
Si	1.12	1360 460	34 11.5	5×10^{22}
Ge	0.66	3900 1900	97.5 47.5	4.4×10^{22}
GaAs	1.42	8000 320	200 8.0	4.42×10^{22}

Metric Conversions			
1 cm = 10,000 μm	1 μm = 1,000 nm	1 cm = 1 x 10^7 nm	1 (μm)^2 = 1 x 10^{-8} cm^2

Carrier Motion		
Diffusion current density (A/cm^2)	$J_{D,p} = -q\,D_p\,\frac{dp}{dx}$ $J_{D,n} = q\,D_n\,\frac{dn}{dx}$	
Drift current density (A/cm^2)	$J_s = q\left(\mu_p\,p + \mu_n\,n\right)E = \frac{E}{\rho}$	Velocities $v_{p,S} = \mu_p E$ $v_{n,S} = -\mu_n E$ $I_{p,S} = v_{p,S} \times p \times q \times A$
Resistivity $(\Omega \cdot cm)$	$\rho = \frac{1}{q\left(\mu_p\,p + \mu_n\,n\right)}$	μ_p μ_n decrease with increase in doping concentration
Microscopic Ohm's Law	$J = \frac{E}{\rho}$	
Einstein Relationship	$V_T = \frac{D_n}{\mu_n} = \frac{D_p}{\mu_p}$	Thermal Voltage $V_T = \frac{k\,T}{q}$ = 25.9mV at 300K = 25mV at 290K
Diode Equations		
Barrier or Built-In Voltage	$V_0 = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right)$	
Width Depletion Region (cm)	$W = \sqrt{\frac{2\varepsilon_S}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)\left(V_0 + V_A\right)}$	$\varepsilon_S = 11.7 \times \varepsilon_o = 1.04 \times 10^{-12}\,F/cm$ Forw Bias: $V_A = -V_F < 0$ Rev Bias: $V_A = V_R > 0$
Depletion Region Depth into n-type	$x_n = \frac{N_A}{N_D} x_p$ $x_n = W\,\frac{N_A}{N_A + N_D}$	Charge Balance $N_A\,q\,A\,x_p = N_D\,q\,A\,x_n$ $W = x_n + x_p$
Reverse Saturation Current (A)	$I_S = q\,A\,n_i^2\left(\frac{D_p}{L_p N_D} + \frac{D_n}{L_n N_A}\right)$	
Shockley Diode Equation (A)	$I_D = I_S\left(e^{V_D/nV_T} - 1\right)$	If n = 1 Ideal Diffusion Typ $1 < n < 2$ n = ideality factor



Periodic Table of the Elements



Diode Capacitance and Diode Analysis Models		
Diode Junction Capacitance (Rev Bias)	$C_{j0} = A\sqrt{\left(\frac{\varepsilon_S q}{2}\right)\left(\frac{N_A N_D}{N_A + N_D}\right)\left(\frac{1}{V_0}\right)}$	$C_j = \frac{C_{j0}}{\left(1+\frac{V_R}{V_0}\right)^m}$ $1/3 \le m \le 1/2$
Ideal Diode Model	ON = Short Circuit: $V_D = 0$ Verify $I_D > 0$	OFF = Open Circuit: $I_D = 0$ Verify $V_D < 0$
Constant Voltage Drop (CVD) Diode Model	ON = Constant Voltage Source: $V_D = 0.7\text{V}$; Verify $I_D > 0$	OFF = Open Circuit: $I_D = 0$ Verify $V_D < 0.7\text{V}$
Shockley Diode Equation (A)	$I_D = I_S\left(e^{V_D/nV_T} - 1\right)$ $\approx I_S e^{V_D/nV_T}$	If n = 1 Ideal Diffusion Typ $1 < n < 2$ n = ideality factor
Gauss-Seidel Update Equations	Load Line $I_{D2} = \frac{V_{th} - kV_{D1}}{R_{th}}$ k: number of diodes in series	Shockley Update $V_{D2} = V_{D1} + n \times V_T \times \ln\left(\frac{I_{D2}}{I_{D1}}\right)$
Diode Small-Signal Model $i_D = I_D + i_d$ $v_D = V_D + v_d$	$i_D = I_D\, e^{v_d/n\times V_T} \approx I_D + \frac{v_d}{r_d}$ $r_d = \frac{n \times V_T}{I_D}$ for $v_d \le n \times 5\text{mV}$	
Line Regulation	$LR = \frac{\Delta V_{out}}{\Delta V_{in}}$	
Zener Diode	$I_Z = \frac{(V_Z - V_{Z0})}{r_z}$	Given a point $V_{Z0} = V_Z - r_z \times I_Z$ $P_Z = I_Z \times V_Z = I_Z \times (V_{Z0} + r_z \times I_Z)$

Diode Labeling:

